NetStream Capture Kit

Collect the evidence we need from your Huawei switches in about one hour.

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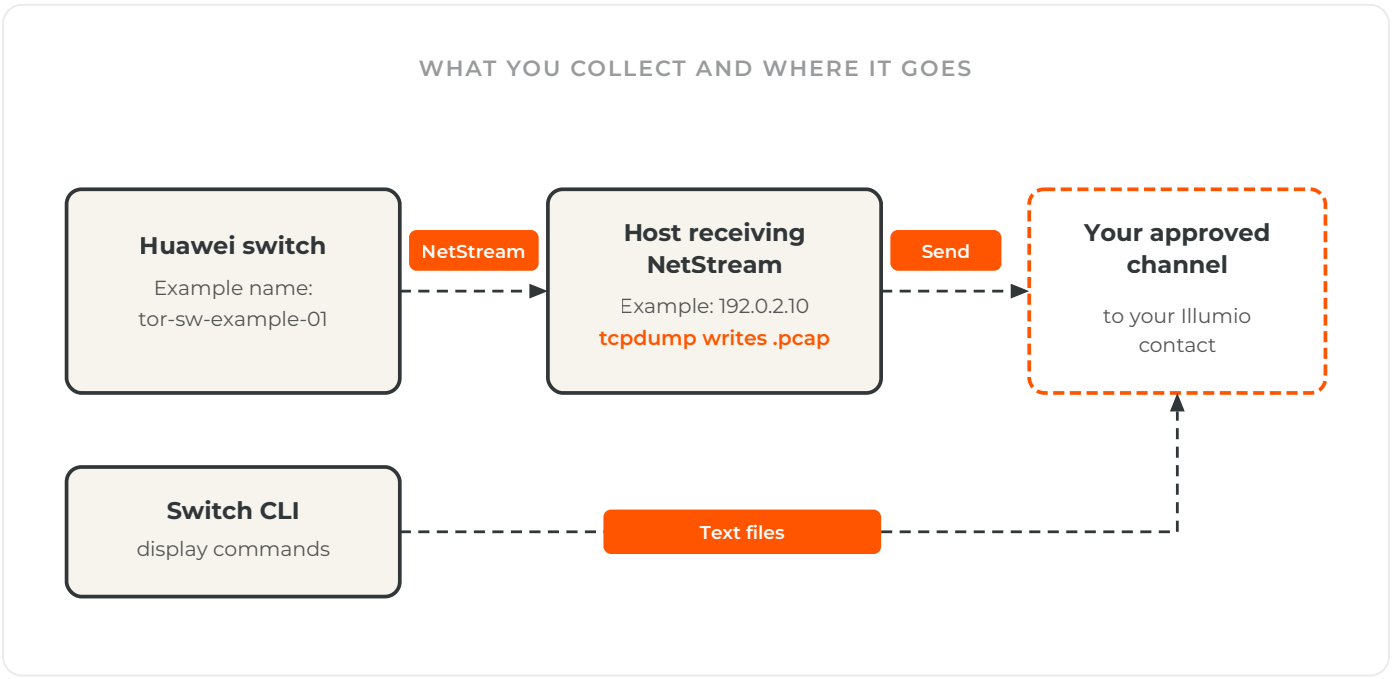
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Overview

To plan the integration of your Huawei switches with Illumio, we need to see what the switches really export. This kit shows you how to collect that evidence with three items: a short network capture, the output of a few `display` commands, and a short questionnaire. Nothing is installed or changed on the switches, except one optional setting described in Part 1.



What the evidence answers

- Whether the switches export **IP flows** (IPv4/IPv6 addresses and ports) or only **Layer 2** (MAC/VLAN) flows.
- Which **NetStream version** each switch uses, and whether IPv6 NetStream is configured.
- Whether a **sampling rate** is configured and whether the flows report it.
- Which **line cards or stack members** export, how often **templates** are refreshed and from which UDP source port.
- Whether the **clock, time zone and NTP** of each device are correct.
- The exact **model and software version** of each switch.

What you will need

- **A Linux host that receives the NetStream packets** (the current collector, or any host that can see the traffic), with `tcpdump` and administrator rights.
- **Read access to each switch** (console or SSH) to run `display` commands.
- **About one hour in total.** The capture runs in the background while you collect the command output.

SOURCES

1. Huawei Encyclopedia — NetStream — info.support.huawei.com/info-finder/encyclopedia/en/NetStream.html
2. Huawei CloudEngine 12800 — NetStream configuration commands — support.huawei.com/enterprise/en/doc/EDOC1100138131/3a962da5/netstream-configuration-commands

Part 1: Capture the NetStream traffic

NetStream is sent by the switch as UDP packets to your collector. Recording those packets on the receiving host shows us exactly what each switch exports. The packets contain flow summaries only (addresses, ports, protocol and counters), not the content of your traffic.

1

Note the values you will use

The network interface of the receiving host that sees the NetStream packets, and the management or source IP address of each switch. Run `date -u` on the host and write down the result: we compare it with the switch clock.

2

Start the capture

Use one file per switch, or one file for all switches (see the second command). Stop with `Ctrl+C`, or use the `timeout` form so that it stops by itself.

CAPTURE HOST (LINUX)

```
# One switch. Replace the values in angle brackets.
sudo tcpdump -i <interface> -s 0 -nn -w netstream.pcap udp and host <switch-ip>

# Several switches in one file, stops by itself after 40 minutes (2400 s).
sudo timeout -s INT 2400 tcpdump -i <interface> -s 0 -nn -w netstream.pcap \
  udp and '(host <switch-1-ip> or host <switch-2-ip>)'
```

3

Make sure the capture includes the templates

Version 9 switches describe their records in a template that is sent only from time to time. Without it the records cannot be read. Choose one option.

OPTION	WHAT TO DO
A. Longer capture (no change on the switch)	Capture for 40 minutes, so the capture covers the default template refresh of 30 minutes[1].
B. Shorter refresh (if your change process allows it)	Set the template refresh to 1 minute during the capture, capture for 15 minutes, then restore your previous value. S-series and AR: `ip netstream export template timeout-rate 1`. CE: `netstream export ip template timeout-rate 1`. Use `?` to confirm the command name on your software release.
C. Capture at export start	If you enable NetStream export or restart it during a maintenance window, start the capture just before. Switches send their templates when export starts.

4

Check the file

List the first packets. You should see UDP packets from the switch IP address to your collector. Many Huawei switches send from UDP source port 40000[1]. If the file is empty, check the interface name, the switch IP address and any firewall between the switch and the host.

CAPTURE HOST (LINUX)

```
tcpdump -nn -r netstream.pcap -c 5
```

NO ACCESS TO THE RECEIVING HOST?

Use port mirroring (SPAN) on the switch port that connects to the collector, and send the mirrored traffic to a Linux host running the same `tcpdump` command on the mirror interface. Use your standard procedure for port mirroring.

SOURCES

1. Huawei NetStream YANG module (template refresh 30 min, source port 40000) — netconfcentral.org/modules/huawei-netstream/2019-06-27
2. tcpdump manual — tcpdump.org/manpages/tcpdump.1.html

Part 2: Collect the switch information

Run the commands below on **each switch** while the capture is running. Save the output of each switch in one text file (for example with your terminal's session log). If a command is not recognized on your software release, type the start of it followed by `?`, use the closest match and continue with the next one.

2.1 Model, software and line cards

SWITCH CLI

```
screen-length 0 temporary
display version
display device
```

2.2 Clock, time zone and NTP

Run `display clock` at the moment you run `date -u` on the capture host, or note both times. Use the NTP command that your software accepts: `ntp-service` on S-series and AR, `ntp` on CE.

SWITCH CLI

```
display clock
display ntp-service status
display ntp-service sessions
# CE software:
display ntp status
display ntp sessions
display current-configuration | include clock
display current-configuration | include ntp
```

2.3 NetStream configuration

SWITCH CLI

```
display current-configuration | include netstream
display current-configuration | include sampler
```

The interface lines (for example `ip netstream inbound` or `netstream inbound ip`) appear under their interface in the full configuration. If you may share the complete configuration, add the output of `display current-configuration`. Otherwise run `display this` inside each interface view that carries NetStream and include the interface name.

2.4 NetStream status and statistics

List the NetStream display commands of your release, then run the ones for statistics and for the flow cache and add their output.

SWITCH CLI

```
display ip netstream ?
# CE software:
display netstream ?
```

What we look for in the configuration

IF THE OUTPUT SHOWS	IT TELLS US
`export version 5` , `9` or `ipfix`	The NetStream version in use
Lines that start with `ipv6 netstream`	IPv6 NetStream is configured
Lines that mention `ethernet`	Layer 2 (Ethernet) flows are configured
`sampler` lines	Sampling mode, rate and direction
`template timeout-rate` and `template option`	Template refresh and option templates
`export source` and `export host`	Export source address, collector address and port
`slot` lines and the `display device` output	Line cards and stack members that export

SOURCES

1. Huawei CloudEngine 12800 — NetStream configuration commands — support.huawei.com/enterprise/en/doc/EDOC1100138131/3a962da5/netstream-configuration-commands
2. Huawei Encyclopedia — NetStream — info.support.huawei.com/info-finder/encyclopedia/en/NetStream.html

Part 3: Questionnaire

Answer in a text file or in your reply, one block per switch. Short answers are enough, and "not known" is a valid answer.

For each switch

#	QUESTION	EXAMPLE ANSWER (FOR ILLUSTRATION)
1	Switch name, management IP address and role	tor-sw-example-01, 198.51.100.11, top of rack
2	Standalone, stack or chassis with line cards?	Stack of two members
3	NetStream version or versions exported (5, 8, 9)	9
4	Is NetStream also configured for IPv6?	No
5	Flow types exported: IP, Ethernet (Layer 2) or both	IP
6	Is sampling configured? Rate and direction (inbound, outbound)	1 in 1000, inbound
7	Interfaces with NetStream enabled	All uplinks
8	Collector address and UDP port, and the VPN instance if any	192.0.2.10, port 2055
9	Were the template refresh or timeout settings changed from the defaults?	No
10	NTP servers in use and the time zone the switch should use	Two servers, UTC-5

General

#	QUESTION	EXAMPLE ANSWER (FOR ILLUSTRATION)
11	Start time, end time and time zone of the capture, and the `date -u` value from the host	2026-11-05 14:00 to 14:40
12	Which collector or tool receives the NetStream today, if any?	Existing flow collector
13	Anything unusual during the capture (maintenance, outage, configuration change)?	None
14	Technical contact for follow-up questions	Your name and email

SOURCES

1. Huawei CloudEngine 12800 — NetStream configuration commands — support.huawei.com/enterprise/en/doc/EDOC1100138131/3a962da5/netstream-configuration-commands

Part 4: Name and send the files

File names

Use this pattern, in lowercase and without spaces: `<site>_<switch>_<content>_<yyyymmdd>.<ext>`. The names below are examples.

FILE (EXAMPLE NAMES)	CONTENT
`site-example_tor-sw-example-01_netstream_20261105.pcap`	Capture from Part 1
`site-example_tor-sw-example-01_display_20261105.txt`	Command output from Part 2
`site-example_questionnaire_20261105.txt`	Answers from Part 3

Package and send

Put all files in one folder, compress it and create a checksum so that we can confirm the file arrived intact.

CAPTURE HOST (LINUX)

```
tar -czf site-example_netstream-kit_20261105.tar.gz site-example/
sha256sum site-example_netstream-kit_20261105.tar.gz
```

Send the archive and the checksum line to the Illumio contact who gave you this kit.

SHARE THROUGH YOUR APPROVED CHANNEL

The capture contains the internal IP addresses of your servers and network devices, and the command output contains device names and addresses. Send the files through the file-sharing channel that your organization approves for this kind of data. The filtered commands in this kit do not output passwords or keys; review the files before sending if you added the full configuration.

Final checklist

- [ ] **Capture file** — one file per switch, or one file with all switches, at least 15 minutes with templates, or 40 minutes.
- [ ] **Command output** — one text file per switch with the commands of Part 2.
- [ ] **Questionnaire** — answers for each switch and the general questions.

- [ ] **Times** — the host `date -u` value and the `display clock` value of each switch.
- [ ] **Package** — archive and checksum sent through the approved channel.

SOURCES

1. GNU Coreutils — sha2 utilities — gnu.org/software/coreutils/manual/html_node/sha2-utilities.html

ABOUT THIS DOCUMENT

This guide was prepared by Eric Roscher – Illumio Pre-Sales, LATAM to guide the collection of NetStream evidence from Huawei switches. It is a working aid for the recipient and is not an official publication of Illumio, Inc.; it does not modify or replace official product documentation, contractual agreements, or statements of work. Validate all technical details against the official documentation for your specific product versions before acting on them. Illumio and the Illumio logo are trademarks of Illumio, Inc., used here to identify the subject matter.

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